equation further

$$p(\mathbf{y}, \mathbf{z}|\mathbf{x}) = \prod_{t \in \mathcal{T}_w} p(w_t, z_t|(y,z)_{<t}, \mathbf{x}) \times \prod_{t' \in \mathcal{T}_l} p(l_{t'}, z_{t'}|(y,z)_{<t'}, \mathbf{x}). \quad (5)$$

Here, $\mathcal{T}_w$ is a set of time steps where $z_t = 1$, and $\mathcal{T}_l$ is a set of time-steps where $z_t = 0$. And, $\mathcal{T}_w \cup \mathcal{T}_l = \{1, 2, \ldots, T_y\}$ and $\mathcal{T}_w \cap \mathcal{T}_l = \emptyset$. We denote all previous observations at step t by $(y,z)_{<t}$. Note also that $\mathbf{h}_t = f((y,z)_{<t})$.

Then, the joint probabilities inside each product can be further factorized as follows:

$$p(w_t, z_t|(y,z)_{<t}) = p(w_t|z_t = 1, (y,z)_{<t}) \times p(z_t = 1|(y,z)_{<t}) \quad (6)$$

$$p(l_t, z_t|(y,z)_{<t}) = p(l_t|z_t = 0, (y,z)_{<t}) \times p(z_t = 0|(y,z)_{<t}) \quad (7)$$

here, we omitted $\mathbf{x}$ which is conditioned on all probabilities in the above.

The switch probability is modeled as a multilayer perceptron with binary output:

$$p(z_t = 1|(y,z)_{<t}, \mathbf{x}) = \sigma(f(\mathbf{x}, \mathbf{h}_{t-1}; \theta)) \quad (8)$$

$$p(z_t = 0|(y,z)_{<t}, \mathbf{x}) = 1 - \sigma(f(\mathbf{x}, \mathbf{h}_{t-1}; \theta)). \quad (9)$$

And $p(w_t|z_t = 1, (y,z)_{<t}, \mathbf{x})$ is the shortlist softmax and $p(l_t|z_t = 0, (y,z)_{<t}, \mathbf{x})$ is the location softmax which can be a pointer network. $\sigma(\cdot)$ stands for the sigmoid function, $\sigma(x) = \frac{1}{\text{exp(-x)+1}}$.

Given N such context and target sequence pairs, our training objective is to maximize the following log likelihood w.r.t. the model parameter θ

$$\arg\max_{\theta} \frac{1}{N} \sum_{n=1}^{N} \log p_\theta(y_n, z_n|x_n). \quad (10)$$

4.1 Basic Components of the Pointer Softmax

In this section, we discuss practical details of the three fundamental components of the pointer softmax. The interactions between these components and the model is depicted in Figure 3.

Location Softmax $\mathbf{l}_t$: The location of the word to copy from source text to the target is predicted by the location softmax $\mathbf{l}_t$. The location softmax outputs the conditional probability distribution $p(l_t|z_t = 0, (y,z)_{<t}, \mathbf{x})$. For models using the attention mechanism such as NMT, we can reuse the probability distributions over the source words in order to predict the location of the word to point. Otherwise we can simply use a pointer network of the model to predict the location.

Shortlist Softmax $\mathbf{w}_t$: The subset of the words in the vocabulary V is being predicted by the shortlist softmax $\mathbf{w}_t$.

Switching network d_t : The switching network d_t is an MLP with sigmoid output function that outputs a scalar probability of switching between $\mathbf{l}_t$ and $\mathbf{w}_t$, and represents the conditional probability distribution $p(z_t|(y,z)_{<t}, \mathbf{x})$. For NMT model, we condition the MLP that outputs the switching probability on the representation of the context of the source text $\mathbf{c}_t$ and the hidden state of the decoder $\mathbf{h}_t$. Note that, during the training, d_t is observed, and thus we do not have to sample.

The output of the pointer softmax, $\mathbf{f}_t$ will be the concatenation of the the two vectors, $d_t \times \mathbf{w}_t$ and $(1 - d_t) \times \mathbf{l}_t$.

At test time, we compute Eqn. (6) and (7) for all shortlist word w_t and all location l_t, and pick the word or location of the highest probability.

5 Experiments

In this section, we provide our main experimental results with the pointer softmax on machine translation and summarization tasks. In our experiments, we have used the same baseline model and just replaced the softmax layer with pointer softmax layer at the language model. We use the Adadelta (Zeiler, 2012) learning rule for the training of NMT models. The code for pointer softmax model is available at `https://github.com/caglar/pointer_softmax`.

5.1 The Rarest Word Detection

We construct a synthetic task and run some preliminary experiments in order to compare the results with the pointer softmax and the regular softmax's performance for the rare-words. The vocabulary size of our synthetic task is $|V| = 600$ using sequences of length 7. The words in the sequences are sampled according to their unigram distribution which has the form of a geometric distribution. The task is to predict the least frequent word in the sequence according to unigram distribution of the words. During the training, the sequences are generated randomly. Before the training, val-